

CIRCUMFERENCE AND AREA OF CIRCLES

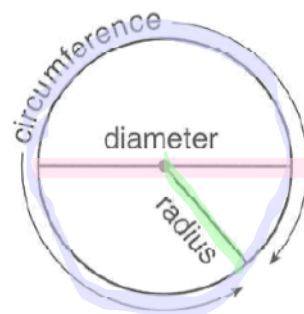
A circle is a set of closed points in a plane. Every point on a circle is the same distance from the center.

The distance around a circle is called the **circumference**. Measuring the circumference of a circle is challenging because you can't use a ruler to measure something round. But you can measure the distance straight across.

The **diameter** is a line segment that passes through the center of a circle. Drawing a diameter cuts the circle into two equal halves.

Half of the diameter is the **radius**. The radius is the distance from the center to any point on the circle.

You can approximate circumference using a relationship that exists between the circumference of a circle and its diameter. Whatever the size of the circle, the circumference is always a little more than 3 times the diameter.



$$C = \pi d$$

$$\pi = C/d$$

if we use 3.14 for π

$$C \approx 3.14 d$$

$$C \approx 3d$$

$$\pi = 3.14 \dots \dots \dots$$

decimal goes on forever

Circumference & Area of a Circle

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$$C = \pi d$$
$$\pi = C/d$$

The ratio of the circumference to the diameter is called pi, represented by the Greek letter π . The value of π is about 3.14. Pi can be used to write formulas for finding the circumference and area of circles. You ~~should~~ **must** memorize these formulas. They will not be listed on the GED math formula sheet.

Circumference (C) of a Circle

$$C = \pi d, \text{ where } d = \text{diameter}$$

Area (A) of a Circle

$$A = \pi r^2, \text{ where } r = \text{radius}$$

USING CIRCLE FORMULAS

Example What is the circumference and area of a circle when PQ equals 12 inches?

PQ is the diameter: $d = 12$

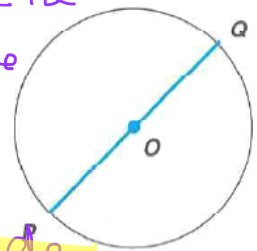
The radius is half the diameter: $r = 6$

Use the circumference formula:

$$C = \pi d$$
$$\approx 3.14 \cdot 12$$
$$\approx 37.68 \text{ inches}$$

Use the area formula:

$$A = \pi r^2$$
$$\approx 3.14 \cdot 6^2$$
$$\approx 3.14 \cdot 36$$
$$\approx 113.04 \text{ square inches}$$



$$C = \pi d$$
$$\approx 3.14(12)$$

$$C \approx 37.68 \text{ inches}$$

Length - one dimension

Remember that area is measured in square units. Even though you can't fit squares into a circle, you can estimate how many would fit.

$$A = \pi r^2$$
$$\approx 3.14(6^2)$$

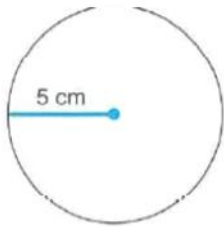
$$A \approx 113.04 \text{ sq in}$$

Examples

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Find the circumference and area of each circle. Round to the nearest tenth.

1.



$$r = 5, d = 10$$

$$C = \pi d$$

$$\approx 3.14(10)$$

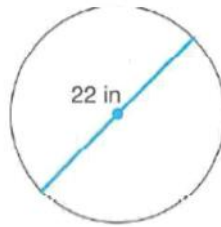
$$C \approx 31.4 \text{ cm}$$

$$A = \pi r^2$$

$$\approx 3.14(5^2)$$

$$A \approx 78.5 \text{ cm}^2$$

2.



$$r = 11, d = 22$$

$$C = \pi d$$

$$\approx 3.14(22)$$

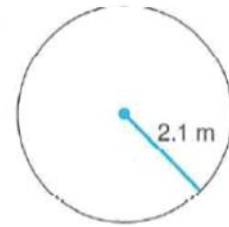
$$C \approx 69.1 \text{ in}$$

$$A = \pi r^2$$

$$\approx 3.14(11^2)$$

$$A \approx 379.9 \text{ in}^2$$

3.



$$r = 2.1$$

$$d = 4.2$$

$$C = \pi d$$

$$\approx 3.14(4.2)$$

$$C \approx 13.2 \text{ m}$$

$$A = \pi r^2$$

$$\approx 3.14(2.1^2)$$

$$A \approx 13.8 \text{ m}^2$$

Examples, cont'd...

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4. Sasha needs to buy one packet of seeds for every 12 square feet in her garden. If the circular garden has a diameter of 18 feet, how many packets of seeds will Sasha need?

$$d = 18, r = 9$$

$$A = \pi r^2$$

$$\approx 3.14 (9^2)$$

$$A \approx 254.34 \text{ ft}^2$$

$$254.34 \text{ ft}^2$$

how many packets do we need?

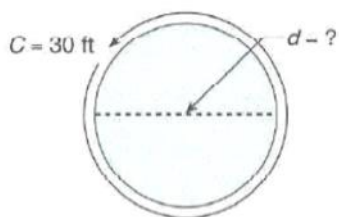
$$\# \text{ packets} = \frac{254.34}{12}$$

$$= 21.195$$

To have a bit more.

$$= 21 \text{ or } 22$$

5. Max is planting a circular flower garden in the center of a lawn. He has 30 feet of fencing to go around the garden. To the nearest tenth of a foot, what is the greatest diameter the garden can have without Max buying more fencing?



6. Cans of frozen juice concentrate often have plastic strips that you pull to remove the lid from the can. The length of the strip is the circumference of the lid plus 0.25 inch. If the radius of the lid is 1.25 inches, what is the length of the plastic strip to the nearest tenth of an inch?



AREA OF COMPLEX FIGURES

A complex figure is made from more than one shape. The area of a complex figure can't be found using a single formula. Instead, you have to develop a plan of your own for finding the area.

Some figures can be cut up into smaller pieces. The area is the sum of the areas of these smaller shapes.

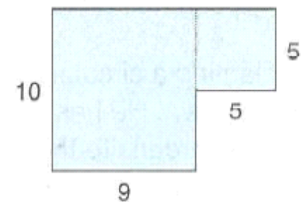
ADDING TO FIND AREA

Example What is the area of the figure in square inches?

Step 1

Divide the figure into smaller parts.

Here, the figure is divided into a rectangle and a square.



Step 2

Find the areas of the parts.

$$\begin{aligned} \text{Rectangle: } A &= 10 \cdot 9 \\ &= 90 \text{ sq in} \end{aligned}$$

$$\begin{aligned} \text{Square: } A &= 5 \cdot 5 \\ &= 25 \text{ sq in} \end{aligned}$$

Step 3

Find the total area.

$$90 + 25 = 115 \text{ sq in}$$

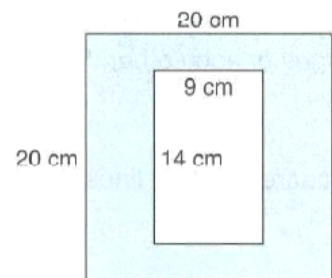
SUBTRACTING TO FIND AREA

Example What is the area of the shaded part of the figure?

Step 1

Divide the figure into smaller parts.

A rectangle is cut out of a square.



Step 2

Find the areas of the parts.

$$\begin{aligned} \text{Square: } A &= 20 \cdot 20 \\ &= 400 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Rectangle: } A &= 14 \cdot 9 \\ &= 126 \text{ cm}^2 \end{aligned}$$

Step 3

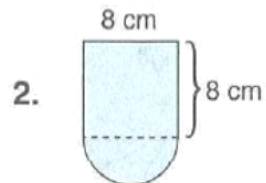
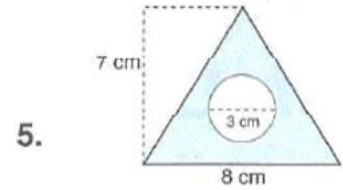
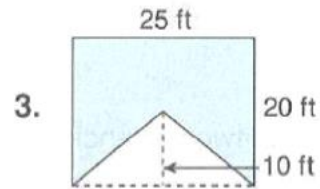
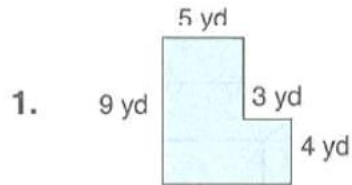
Find the area of the shaded portion.

$$400 - 126 = 274 \text{ cm}^2$$

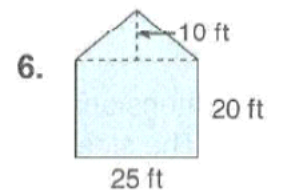
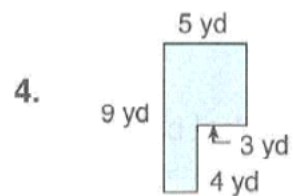
More Examples

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Find the area of each figure.



(*Hint:* This figure contains a half-circle.)

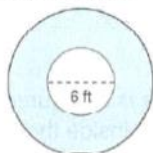


More Examples, cont'd...

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Solve.

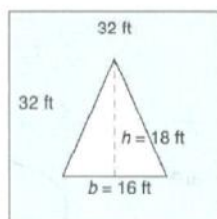
7. A circular fountain has a 3-foot walkway surrounding it. What is the area of the walkway? Round to the nearest tenth.



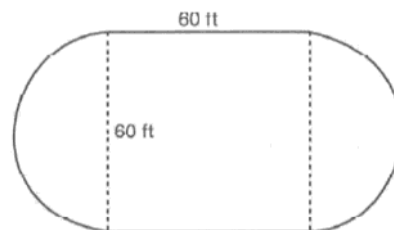
8. An X is formed by drawing one 11-by-6-inch rectangle on top of another. What is the area in square inches of the X?



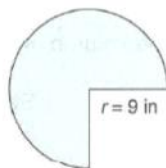
9. The shaded area represents a walkway around a triangular garden. What is the area of the walkway?



10. An event planner has the following space to create a dirt bike course. To the nearest square foot, what is the area of the space?



11. **Investigate** The figure below is $\frac{3}{4}$ of a circle.



- a. Find the area of the figure.
- b. If a square is drawn with the same area, what would a side of the square measure to the nearest inch?